

1. Suppose that 15 points are selected at random and independently from the interval $(0, 1)$. In average, how many of them will be greater than $3/4$?

$$P(X) = \frac{1 - 3/4}{1 - 0} = \frac{1}{4}$$

$$X \sim B(15, \frac{1}{4})$$

$$E(X) = np = 15 \times \frac{1}{4} = \frac{15}{4}$$

ANS: $\frac{15}{4}$

2. A line segment of length k is cut randomly into two segments. What is the probability that none of the two segments is smaller than $k/3$?

$$P(X) = \frac{\frac{1}{3}k}{k} = \frac{1}{3}$$

ANS: $\frac{1}{3}$

3. Let X be a uniform random number from $(0, 1)$. Find the density functions of $Y = -\ln(1 - X)$.

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x \end{cases}$$

$$G(t) = P(Y < t) = P(-\ln(1-x) < t)$$

$$t \in (0, \infty)$$

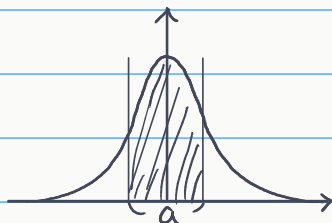
$$= P(\ln(1-x) > -t) = P(1-x > e^{-t})$$

$$= P(x < 1 - e^{-t})$$

$$= \begin{cases} 0 & t \leq 0 \\ 1 - e^{-t} & t > 0 \end{cases}$$

$$\text{ANS: } g(t) = \begin{cases} 0 & t \leq 0 \\ e^{-t} & t > 0 \end{cases}$$

4. Let Z be a standard normal random variable and α be a given positive constant. Find the real number x that maximizes $P(x < Z < x + \alpha)$.

ANS: $-\frac{\alpha}{2}$

5. The grades for a certain exam are normally distributed with mean 67 and variance 64. What percent of students get A(≥ 90), B(80 - 90), C(70 - 80), D(60 - 70), and F(< 60)?

$$\mu = 67 \quad \sigma^2 = 64 \quad \sigma = 8$$

$$A: 1 - P(X \leq 90)$$

$$= 1 - P\left(\frac{X - \mu}{\sigma} \leq \frac{90 - 67}{8}\right)$$

$$= 1 - \Phi\left(\frac{23}{8}\right) \approx 0.00212 \%$$

$$B: P\left(\frac{80 - 67}{8} \leq Z \leq \frac{90 - 67}{8}\right)$$

$$= \Phi\left(\frac{23}{8}\right) - \Phi\left(\frac{13}{8}\right)$$

$$= 0.99788 - 0.94738 = 0.0505 \%$$

$$C: P\left(\frac{70 - 67}{8} \leq Z \leq \frac{80 - 67}{8}\right)$$

$$= \Phi\left(\frac{13}{8}\right) - \Phi\left(\frac{3}{8}\right)$$

$$= 0.94738 - 0.64431 = 0.30307 \%$$

$$D: P\left(\frac{60 - 67}{8} \leq Z \leq \frac{70 - 67}{8}\right)$$

$$= \Phi\left(\frac{3}{8}\right) - \Phi\left(-\frac{7}{8}\right)$$

$$= 0.64431 - (1 - 0.80785) = 0.45216 \%$$

$$E: P\left(Z \leq \frac{60 - 67}{8}\right)$$

$$= \Phi\left(-\frac{7}{8}\right)$$

$$= 1 - 0.80785 = 0.19215 \%$$

6. Let $X \sim N(\mu, \sigma^2)$. Prove that $P(|X - \mu| > k\sigma)$ does not depend on μ or σ .

Since μ is the average value of X , so subtract it would cause the graph to be centered to $x=0$, then divide X by σ would turn standard deviation into 1, so these two steps would turn all Normal R.V. into a Standard Normal R.V., thus the probability would be the same since it's already standardized

$$P(|X - \mu| > k\sigma) = P\left(\frac{|X - \mu|}{\sigma} > k\right) = P(|Z| > k)$$

$$= P(Z > k) + P(Z < -k)$$

$$= 1 - \Phi(k) + (1 - \Phi(k))$$

$$= 2 - 2\Phi(k)$$

↑ not related to μ & σ

7. Let $X \sim N(0,1)$. Calculate the probability density function of $Y = \sqrt{|X|}$.

$$F(x) = \Phi(x)$$

$$G(t) = P(Y < t) = P(\sqrt{|X|} < t) = P(-t^2 < X < t^2)$$

$$= \Phi(t^2) - \Phi(-t^2)$$

$$= 2\Phi(t^2) - 1$$

$$g(t) = G'(t) = 2\phi(t^2) \cdot 2t = \begin{cases} 2\phi(t^2) \cdot 2t & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

8. Let X be an exponential random variable with parameter λ , mean $E(X)$ and standard deviation σ_X . Find $P(|X - E(X)| > 2\sigma_X)$.

$$E(X) = \frac{1}{\lambda} \quad \text{Var}(X) = \frac{1}{\lambda^2} \quad \sigma = \frac{1}{\lambda}$$

$$P(|X - E(X)| > 2\sigma_X) = P(X < -2\sigma_X + E(X)) + P(X > 2\sigma_X + E(X))$$

$$= F(-\frac{1}{\lambda}) + (1 - F(\frac{3}{\lambda}))$$

$$= 0 + 1 - (1 - e^{-3}) = e^{-3}$$

9. Let X , the lifetime (in years) of a radio tube, be exponentially distributed with mean $1/\lambda$. Find the probability mass function (p.m.f.) of $\lfloor X+1 \rfloor$. (It is noted that $\lfloor X+1 \rfloor$ is the integer part of $X+1$, i.e. the greatest integer less than or equal to $X+1$. Actually, $\lfloor X+1 \rfloor$ can be proved to be a Geometric random variable.)

$$P(Y=k) = P(\lfloor X+1 \rfloor = k)$$

$$= P(k \leq X+1 < k+1)$$

$$= P(k-1 \leq X < k)$$

$$= F(k) - F(k-1)$$

$$= 1 - e^{-\lambda k} - (1 - e^{-\lambda(k-1)})$$

$$= 1 - e^{-\lambda k} - 1 + e^{-\lambda k} \cdot e^{\lambda}$$

$$= (e^{\lambda} - 1) e^{-\lambda k}$$

$$= (e^{\lambda} - 1) e^{-\lambda(k-1)} \cdot e^{-\lambda}$$

$$= (1 - e^{-\lambda}) \cdot e^{-\lambda(k-1)}$$

$$= (1-p) \cdot p^{k-1} \quad \dots \quad p = e^{-\lambda}$$